Disentangling Semantic Knowledge and Memorisation in LLM-Based Data Cleaning: A Benchmark on Synthetic vs. Real Tabular Data

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Recap: Problem and Research Questions

- Papers report that LLMs fill missing values well, but they test on famous datasets the model may have seen during training.
- A good score can then mean **understanding** (the model knows what the columns mean) or **memory** (it recalls the records).
- We built a benchmark that changes one thing at a time and measures how the LLM's error moves.

No.	Question
RQ1	How much of the LLM's skill comes from column meaning, and how much from remembered records?
RQ2	If a model has memorised a dataset more, does it impute it better?
RQ3	Does an LLM handle MNAR suspiciously well on famous data, but not on a synthetic copy?

Table 1: Research questions answered in this work (RQ4 and RQ5 remain future work)

Research Design

Nine versions of every dataset

Factor	Level	What it removes
Origin tests memory	real	nothing (baseline)
	perturbed	exact text of the published file
	synthetic twin	every record that could be recalled
Naming tests meaning	real	nothing (baseline)
	hidden (f1, f2, …)	what the columns mean
	shuffled	meaning, plus a misleading cue

Table 2: 3 origins × 3 namings = 9 versions; real and perturbed copies are scored on the same cells

The effects we measure

$$\Delta_{\text{mem}} = [\text{E}(\text{twin}) - \text{E}(\text{real})] - [\text{E}_{\text{ref}}(\text{twin}) - \text{E}_{\text{ref}}(\text{real})]$$

Extra error when nothing can be recalled. The classical reference (MICE) removes any difference in difficulty. **Positive = memory.**

$$\Delta_{\text{sem}} = \text{E}(\text{hidden}) - \text{E}(\text{real names})$$

How much the model relies on column meaning; Δ_{mislead} is the same with shuffled names. Classical methods never see names.

$$\text{D} = \text{E}(\text{MNAR}) - \text{E}(\text{MCAR})$$

Recall would make D suspiciously small on the real table only (the MNAR fingerprint).

Research Design (Contd.)

A fair synthetic twin

Twin generator	Difficulty gap
Gaussian copula per class	**0.046**
Gaussian copula (whole table)	0.087
TVAE	0.133
CTGAN	0.235

Table 3: Mean change in classical error between real table and twin (lower = fairer). The per-class copula keeps class clusters such as Iris's three species.

Masks, scores and statistics

- **Missingness:** MCAR, MNAR (large values hidden more often) and Pima's real pattern, at exact rates.
- **Scores:** NRMSE (error ÷ column SD) and the robust *nmdae* (median error), which a few wild answers cannot inflate.
- **Paired:** every method is scored on the same hidden cells.
- **Uncertainty:** every prediction is saved; 2,000 bootstrap redraws of the scored rows give 95% intervals; Holm's correction controls false alarms.
- **Direct memory test:** Bordt et al.'s header, row and feature tests on each dataset's CSV file.

Experimental Setup

Datasets	Iris, Wine, Breast Cancer, Pima (famous) and DiaBD 2025 (unseen control)	**Language models**	Qwen2.5 3B / 7B / 14B, Llama 3.1 8B, Gemma 2 9B (4-bit)
Classical baselines	Mean, median, k-NN, MICE, MissForest	**Prompting**	Zero-shot and 4-example few-shot, JSON schema, temperature 0
Hardware	Google Colab free tier, NVIDIA T4 (15 GB)	**Software**	Python, pandas, scikit-learn, SDV, Ollama, tabmemcheck

14,650
LLM requests

0
unusable answers

428
LLM runs (incl. a 99-run pilot), each one version of a dataset

4.2 h
of T4 GPU time

Experiments

Experiment	Models	Prompts	Missingness	Datasets	Rows scored
A. Cross-model comparison	all five	zero-shot	MCAR 20%	all five	20 per run
B. Prompting and MNAR	Qwen2.5-7B	zero- and few-shot	MCAR, MNAR 20%	Iris, Wine, Breast Cancer	50 per run
C. Memorisation test	all five	–	–	all five	10–25 tries
Reference	mean, k-NN, MICE	–	as above	all five	same cells

Table 4: Experiments. Every language-model run covers all nine versions of a dataset; seed 0.

> Each experiment changes one thing: A compares models, B compares prompting styles and missingness, and C measures memory directly, independent of imputation.

Results: How Accurate Are the LLMs?

Dataset	Mean	k-NN	MICE	Qwen2.5-3B	Qwen2.5-7B	Llama 3.1 8B	Gemma 2 9B	Qwen2.5-14B
Iris	0.95	0.46	**0.30**	0.69	0.59	1.03	0.87	0.53
Wine	0.83	**0.79**	0.81	1.00	1.04	1.16	0.95	0.93
Breast Cancer	0.82	0.41	**0.21**	4.23	1.45	1.41	1.03	0.88
Pima	1.14	1.14	**0.89**	1.41	1.36	1.26	1.26	1.35
DiaBD	0.77	0.85	**0.75**	5.00	0.91	0.86	4.20	0.85

Table 5: NRMSE on the real table with real names, 20% MCAR, zero-shot (lower is better; bold = best; red = worse than filling in the column mean)

> MICE is the most accurate method on 4 of 5 datasets (k-NN on Wine). The language models beat the column mean only on Iris, the best-known dataset. Within the Qwen family the 14B model is the most accurate on every dataset, but still behind MICE.

Results: Reliance on Column Names (RQ1)

Shuffled names: $\Delta_{mislead}$

Dataset	Qwen2.5-3B	Qwen2.5-7B	Llama 3.1 8B	Gemma 2 9B	Qwen2.5-14B
Iris	+2.30★	+1.06★	+0.09	+1.04★	+2.04★
Wine	+0.97	+0.10	−0.05	−0.01	+0.12
Breast Cancer	+60.16	+13.74	+0.34	+0.64★	+0.34
Pima	+1.36★	+0.78★	+0.49	+0.45	+0.93★
DiaBD	+5.69★	+3.20★	+2.69★	+4.19	+6.16★

Hidden names: Δ_{sem}

Dataset	Qwen2.5-3B	Qwen2.5-7B	Llama 3.1 8B	Gemma 2 9B	Qwen2.5-14B
Iris	+0.49	+0.43	+0.00	+0.15	+0.67
Wine	+0.27	−0.09	+0.06	−0.01	−0.08
Breast Cancer	+1.27	+0.11	+0.94★	+0.26	+0.42★
Pima	+0.75	+0.48★	+0.77★	+0.57★	+0.34
DiaBD	+1.84	+2.25★	+3.23★	+1.16	+2.33★

Table 6: Change in NRMSE when names are shuffled or hidden (positive = worse without the real names). ★ = significant after Holm's correction.

> 20 of the 50 tests show significant harm when the names are taken away, and none shows a significant gain. The exception is Wine, whose chemical column names carry little meaning for the models.

Results: What the Models Have Memorised

Model	Iris	Wine	Breast Cancer	Pima	DiaBD
Qwen2.5-3B	0/10 (17)	0/10 (3)	0/10 (3)	0/10 (1)	0/10 (3)
Qwen2.5-7B	**11/25** (26)	0/25 (7)	0/25 (3)	0/25 (8)	0/25 (4)
Llama 3.1 8B	0/10 (190)	0/10 (3)	0/10 (3)	0/10 (4)	0/10 (3)
Gemma 2 9B	**2/10** (92)	0/10 (3)	0/10 (3)	0/10 (4)	0/10 (3)
Qwen2.5-14B	**6/10** (404)	0/10 (52)	0/10 (5)	0/10 (8)	0/10 (3)

Table 7: Row test: rows reproduced exactly / tries (in brackets: characters of the file's opening continued exactly). Bordt et al.'s tests; DiaBD (2025) is the control.

> Only Iris has been memorised: Qwen2.5-14B recites 6 of 10 rows and Qwen2.5-7B 11 of 25, Gemma 2 of 10. No model recites a single row of Wine, Breast Cancer or Pima, and DiaBD (the control) stays at zero, as it must.

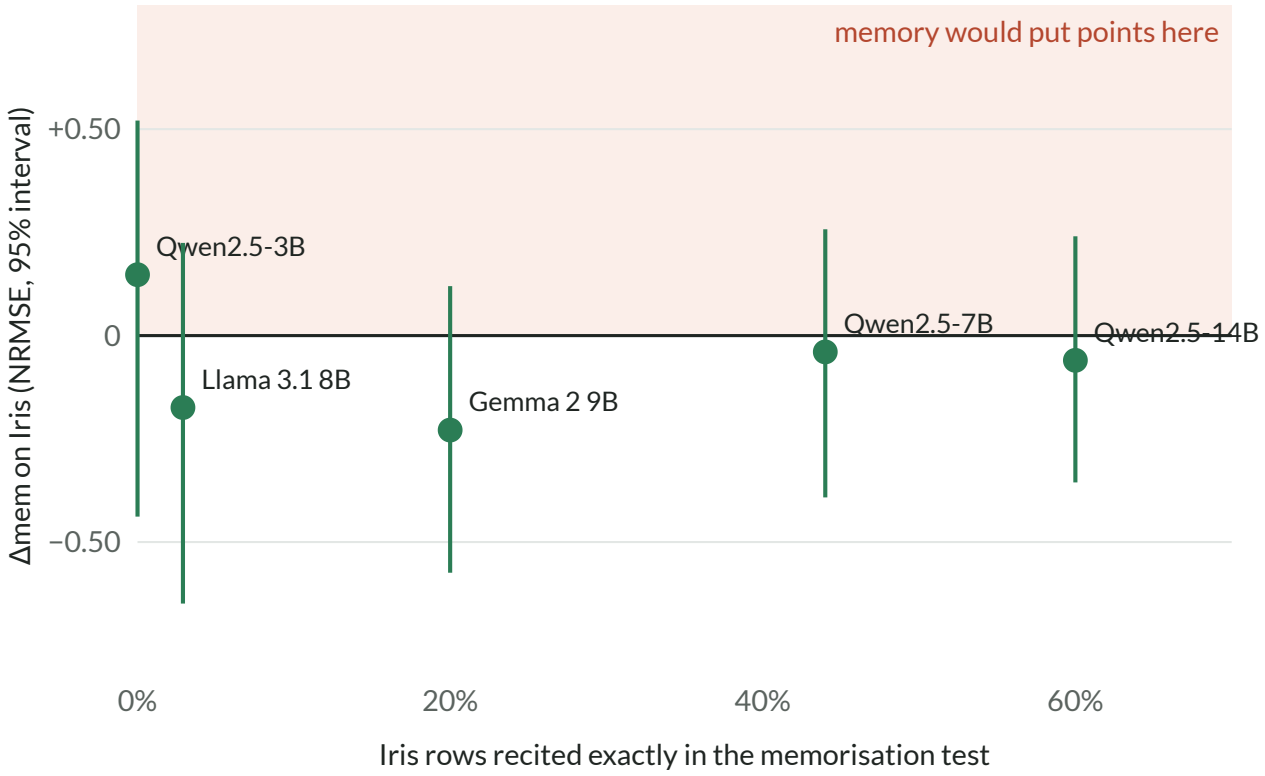
Results: Does Memory Help Imputation? (RQ1)

Dataset	Qwen2.5-3B	Qwen2.5-7B	Llama 3.1 8B	Gemma 2 9B	Qwen2.5-14B
Iris	+0.15 [−0.44, +0.52]	−0.04 [−0.39, +0.26]	−0.17 [−0.65, +0.22]	−0.23 [−0.57, +0.12]	−0.06 [−0.36, +0.24]
Wine	+0.61 [−0.03, +1.09]	+0.13 [−0.21, +0.44]	+0.13 [−0.17, +0.44]	+0.05 [−0.29, +0.32]	+0.11 [−0.19, +0.40]
Breast Cancer	−0.62 [−2.56, +1.66]	+0.05 [−0.33, +0.40]	+0.12 [−0.21, +0.41]	−0.10 [−0.40, +0.17]	+0.09 [−0.16, +0.37]
Pima	−0.25 [−0.58, +0.17]	−0.37 [−0.58, +0.02]	+0.26 [−0.25, +0.61]	−0.28 [−0.52, +0.02]	−0.39 [−0.65, −0.05]
DiaBD	−2.19 [−3.92, −0.95]	+0.28 [−0.01, +0.56]	+0.30 [+0.07, +0.59]	−3.02★ [−3.75, −2.76]	+0.24 [−0.03, +0.54]

Table 8: Δ_{mem} (NRMSE) with 95% bootstrap intervals; reference MICE. Positive = the model did worse once nothing could be recalled. ★ = significant after Holm's correction.

> No Δ_{mem} is significantly above zero for any model or dataset. The only significant value is negative (Gemma on DiaBD, where its answers on the real table broke down). Replacing every record with a twin does not make the models worse.

Results: More Memory, Better Imputation? (RQ2)



Memory does not turn into accuracy

- On Iris, the models range from **0%** to **60%** of rows recited.
- Their Δ_{mem} on Iris lies between −0.23 and +0.15, and every interval includes zero.
- Qwen2.5-14B, the model that remembers Iris best, shows Δ_{mem} = −0.06 [−0.36, +0.24].
- **Answer to RQ2:** more memorisation did not mean better imputation.

Results: Examples in the Prompt (Experiment B)

Dataset	Missingness	Zero-shot	Few-shot	Mean	MICE
Iris	MCAR	0.44	**0.31**	0.67	**0.21**
Iris	MNAR	0.91	**0.72**	1.09	**0.42**
Wine	MCAR	0.78	**0.64**	0.64	**0.45**
Wine	MNAR	1.80	**1.46**	1.55	**0.93**
Breast Cancer	MCAR	1.49	**0.57**	0.77	**0.22**
Breast Cancer	MNAR	1.33	1.57	1.75	**0.56**

Table 9: Typical error (nmdae) of Qwen2.5-7B on the real table with real names, 50 scored rows; MICE for reference

- Four example rows lower the typical error in **5 of 6** settings, most on Breast Cancer under MCAR (1.49 → 0.57).
- Examples shrink the damage from shuffled names wherever it occurred (5 of 6 settings), e.g. Breast Cancer under MCAR **+16.9 → +3.6** and Iris **+0.76 → +0.38**.
- With rows in front of it, the model reads each column's scale from the data instead of trusting the label.
- MICE remains more accurate.

Results: Missing Not At Random (RQ3)

Dataset	Prompt	D LLM real	D LLM twin	D MICE real	D MICE twin	Fingerprint
Breast Cancer	few-shot	0.76	0.85	0.69	0.94	**−0.16**
Breast Cancer	zero-shot	−0.32	−0.34	0.69	0.94	**−0.27**
Iris	few-shot	0.79	0.37	0.39	0.53	**−0.55**
Iris	zero-shot	0.62	0.68	0.39	0.53	**−0.07**
Wine	few-shot	0.65	0.73	0.34	0.45	**−0.03**
Wine	zero-shot	0.75	0.85	0.34	0.45	**−0.01**

Table 10: D = error under MNAR minus error under MCAR (NRMSE), for Qwen2.5-7B and MICE. Fingerprint = $[D_{twin} - D_{real}]_{LLM} - [D_{twin} - D_{real}]_{MICE}$; recall would make it positive.

> Hiding large values (MNAR) sharply raises the error of the classical methods (MICE: Iris 0.36 → 0.75, Breast Cancer 0.31 → 0.99). If the model were recalling values, it would cope suspiciously well on the real table, giving a positive fingerprint. All six fingerprints are zero or negative: **no MNAR memory signature (RQ3)**.

Key Findings

What the models rely on

01 **Column meaning.** Hiding or shuffling names hurts significantly in 20 of 50 tests, never helps.

02 **Examples.** Four example rows cut the typical error in 5 of 6 settings and blunt misleading names.

03 **Scale.** Qwen2.5-14B is the most accurate Qwen on every dataset.

What they do not rely on

01 **Memory.** Iris is memorised (up to 60% of rows), yet no Δ_{mem} is significantly positive.

02 **MNAR recall.** No positive MNAR fingerprint.

03 **Beating MICE.** Open models up to 14B, zero-shot, stay behind MICE and mostly behind the column mean.

Conclusion

RQ	Answer
RQ1	The models' skill comes from **column meaning**, not remembered records: names matter significantly in 20 of 50 tests, while removing every record (synthetic twin) never makes them significantly worse.
RQ2	**No.** Memorisation of Iris ranges from 0% to 60% of rows across models, but Δ_{mem} on Iris stays at zero for all of them.
RQ3	**No.** The MNAR fingerprint is zero or negative everywhere: no sign of recalled values under MNAR.

> For practice: high LLM imputation scores on famous datasets should not be dismissed as memorisation for open models up to 14B, but they are also not yet better than MICE. The benchmark (nine versions, paired scoring, bootstrap intervals) can test any new model the same way.

Limitations and Future Work

Limitations

- One random seed; 20–50 scored rows per run.
- Open models up to 14B parameters, 4-bit, run locally.
- Identical runs can differ by up to about 0.1 NRMSE (batched inference), so small differences are not interpreted.
- The cross-model comparison uses zero-shot prompts and MCAR only.

FUTURE WORK, WITH OUR EXPECTATIONS (HYPOTHESES, NOT RESULTS)

- **Few-shot for all models:** expected to cut the shuffled-name damage by more than half, as for Qwen2.5-7B.
- **Step-by-step prompting** (Abate et al.): expected to help less than examples, at about 3× the cost.
- **MAR and 10% / 30% missingness:** MAR expected close to MCAR; MNAR errors expected to grow with the rate.
- **More seeds and repeats:** expected to tighten the estimates without changing the "no memory advantage" result.
- **Downstream accuracy** (RQ4/RQ5 and classifiers on cleaned data): differences under 2 points expected at 20% missingness.

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Thank You

Questions are welcome